Emily Savoie

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Citizenship: United States of America

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Research interests

Community ecology, eDNA, marine ecology, hypoxia, ecosystem resilience, benthic ecology

Education

2014 – 2018 **Louisiana State University** – Baton Rouge, Louisiana
BS in Biological Sciences
Concentration: Marine Biology
BS in Microbiology
Mentor: Dr. Cameron Thrash *GPA: 4.00*

2010 – 2014 **Opelousas High School** – Opelousas, Louisiana
HS Diploma
GPA: 4.00

Research experience

February 2023 – Present **Research Associate**
Marine Community Ecology Lab (Louisiana State University Department of Oceanography and Coastal Sciences)

- Managing undergraduate students and their projects
- Assisting graduate students with projects
- Teaching new lab personnel protocols and procedures
- Maintaining laboratory inventory and ordering of supplies
- Maintaining care of live animals in the laboratory
- Troubleshooting laboratory equipment
- Running experiments for preliminary data
- Planning, managing, and assisting with field work
- Benthic invertebrate species identification
- Project and data management

October 2021 – February 2023

Research Associate

Abdelmoneim Lab (Louisiana State University School of Veterinary Medicine)

- Managing undergraduate students and their projects
- Assisting graduate students with projects
- Teaching new lab personnel protocols and procedures
- Maintaining laboratory inventory and ordering of supplies
- Maintaining care of live animals in the laboratory
- Troubleshooting laboratory equipment
- Running experiments for preliminary data

January 2019 – October 2021

Research Associate

Geaux Viral Lab (Louisiana State University Department of Oceanography and Coastal Sciences)

- Managing undergraduate students and their projects
- Assisting graduate students with projects
- Teaching new lab personnel protocols and procedures
- Maintaining laboratory inventory and ordering of supplies
- Maintaining care of live animals in the laboratory
- Troubleshooting laboratory equipment
- Running experiments for preliminary data
- Writing protocols for laboratory procedures

December 2017 – August 2018

Undergraduate Research Assistant

Chakrabarty Lab (Louisiana State University Museum of Natural Science)

- Project entitled ”Is Peter Wainwright Right?”
- Researched various characteristics of Acanthomorph fish families
- Researched and updated information and status of several species on the International Union for Conservation of Nature (IUCN)

August 2014 – December 2018

Undergraduate Research Assistant

Thrash Lab (Louisiana State University Department of Biological Sciences)

- Project entitled ”Ecophysiology of the Cosmopolitan OM252 Bacterioplankton (Gammaproteobacteria)”
- Performed physiological experiments to determine potential ecological impacts of bacterial isolate from the northern Gulf of Mexico
- Analyzed genomic data to infer possible metabolic pathways being used by a bacterial isolate from the northern Gulf of Mexico

Publications

2025 **Early developmental exposure to lead (Pb) as a risk factor for stress-related disorders investigated in larval zebrafish (Danio rerio)**
Wilson, M. R., **Savoie, E. R.**, Christofferson, R., Abdelmoneim, A.
Toxicological Sciences, 205(2), 344-357.

2022 **A CURE for physiological characterization of bacterioplankton in liquid culture**
Lanclos, V. C., Coelho, J. T., Hyer, A. J., Brooks, M. M., **Savoie, E. R.**, Kosiba, S., Thrash, J. C.
Journal of Microbiology and Biology Education. 23(2): e00068-22.

2021 **Microbial ecology of oxygen minimum zones amidst global ocean deoxygenation**
Long, A. M., Jurgensen, S. K., Petchel, A. R., **Savoie, E. R.**, Brum, J. R.
Frontiers in Microbiology. 3272.

2021 **Ecophysiology of the cosmopolitan OM252 bacterioplankton (Gammaproteobacteria)**
Savoie, E. R., Lanclos, V. C., Henson, M. W., Cheng, C., Getz, E. W., Barnes, S. J., … & Thrash, J. C.
mSystems. 6(3): e00276-21.

In review or near submission

Effects of riverine input and hypoxia on viral and bacterial communities in the northern Gulf of Mexico
Jurgensen, S. K., Krauss, C. L., **Savoie, E. R.**, Petchel, A. R., Tupitza, J. C., Glaspie, C. N., and Rabalais, N. N.

Presentations

* = Presenting author

November 2025 Louisiana Deltaic Estuaries MBON: Sea Level Rise Sentinels
Savoie, E., Glaspie, C.
Oral presentation, Coastal And Estuarine Research Federation Conference, Richmond, VA

April 2025 — Impacts of Multi-year Drought on Benthic Macrofauna Diversity in Lake Borgne
Savoie, E., Glaspie, C.
Oral presentation, Benthic Ecology Meeting, Mobile, AL

April 2024 — Louisiana Deltaic Estuaries MBON: Sea Level Rise Sentinels
Savoie, E., Tupitza, J., Glaspie, C.
Oral presentation, Benthic Ecology Meeting, Charleston, SC

November 2023 — Louisiana Deltaic Estuaries MBON: Sea Level Rise Sentinels
Savoie, E., Tupitza, J., Glaspie, C.
Poster presentation, Coastal And Estuarine Research Federation Conference, Portland, OR

October 2022 — Toward High-Throughput Screening of Chemicals for Developmental Defects and Behavioural Alterations using the Zebrafish Model
Savoie, E., McAtee, D., Edwards, B., *Wilson, M., Abdelmoneim, A. *Poster presentation, South Central Society of Toxicology*

July 2018 — Is Peter Wainwright Right?
Nall, E., Ludt, B., Faircloth, B., Chakrabarty, P.
Poster presentation, Joint Meeting of Ichthyologists and Herpetologists, Rochester, NY

April 2018 — Metabolic and Physiological Flexibility from an Isolate from the OM252 Clade
Nall, E., Henson, M., Lanclos, V. C., Thrash, J. C.
Poster presentation, Louisiana State University Discover Day, Baton Rouge, LA

June 2017 — Metabolic and Physiological Flexibility in a Coastal Isolate From the OM252 Clade of *Gammaproteobacteria*
Nall, E., Henson, M., Lanclos, V. C., Thrash, J. C.
Poster presentation, American Society for Microbiology Microbe, New Orleans, LA

April 2017 — Physiological and Genomic Characterization of Novel Coastal Isolate of OM252 Clade
Nall, E., Henson, M., Lanclos, V. C., Thrash, J. C.
Poster presentation, Louisiana State University Discover Day, Baton Rouge, LA

February 2016 — Towards Defining the Ecological Niches of Novel Coastal Gulf of Mexico Bacterial Isolates
*Henson, M. W., **Nall, E.**, Pitre, D. M., Weckhorst, J. L., Lanclos, V. C., Thrash, J. C.
Name of conference, workshop, seminar, etc., or a description

April 2016 — Characterization of a Novel Bacterial Isolate From Louisiana Coastal Waters
Nall, E., Henson, M., Lanclos, V. C., Thrash, J. C.
Name of conference, workshop, seminar, etc., or a description

Awards and grants

2017 Undergraduate Research Travel Award $600 (Louisiana State University)

2017 Discover Travel Stipend $1,000 (Louisiana State University)

2017 Fall President's Honor Roll for 4.0 (Louisiana State University)

2017 Spring President's Honor Roll for 4.0 (Louisiana State University)

2016 Spring Dean's List for above 3.5 (Louisiana State University)

2016 Fall Dean's List for above 3.5 (Louisiana State University)

2016 Michael Ippolito Scholarship $1,000 (Louisiana State University)

2015 Undergraduate Research Opportunities Program $2,500 (Louisiana State University Sea Grant)
Project entitled "Characterization of a Novel Bacterial Isolate from Louisiana Coastal Waters with the Potential for Hydrocarbon Degradation".

2015 Spring President's Honor Roll for 4.0 (Louisiana State University)

2014 Fall President's Honor Roll for 4.0 (Louisiana State University)

2014 Valedictorian (Opelousas High School)

Technical skills

Programming languages
Proficient in: R
Familiar with: Python

Software
Proficient in: LaTeX, ImageJ, Microsoft Excel, Knowledge Network for Biocomplexity (KNB)
Familiar with: Git, International Union for Conservation of Nature (IUCN), GraphPad, Joint Genome Institute database

Certifications
- AAUS Scientific Diver
- DAN Diving First Aid (DFA)
- Air Fill Station Operator

Quantitative Skills and Equipment

- Polymerase Chain Reaction (PCR)
- Gel electrophoresis and imaging
- Epifluorescence microscopy
- High-throughput dilution-to-extinction culturing methods
- Interpretation of TEM images of environmental samples
- DNA extraction from environmental samples
- Benthic invertebrate taxonomic identification
- Qubit DNA Quantification
- Guava benchtop flow cytometer
- FilterMax F5 Multi-Mode Microplate Reader
- ViewPoint ZebraBox for behaviorial assays
- OpenTrons OT-2 automated liquid handling robot